THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

DATA SCIENCE • PROJECT SYNOPSIS

Title: Data-Driven Analysis & Forecasting of Renewable and Non-Renewable Electricity Generation in India

INTRODUCTION

India's growing electricity demand makes a balanced and sustainable energy mix increasingly important. This project applies Data Science and Machine Learning to study electricity-generation patterns, compare renewable and non-renewable sources, identify regional trends, and forecast future generation.

PROBLEM STATEMENT : Develop a data-driven system to analyze, compare, visualize, and forecast electricity generation across Indian states and energy sources.

OBJECTIVES

- Analyze electricity generation by state, source, and year.
- Compare renewable and non-renewable contributions to the energy mix.
- Identify growth trends and regional patterns in electricity generation.
- Apply Machine Learning / time-series methods to forecast future generation.

DATASETS INFO:- This project includes various datasets from the official website of CKAN India data portal. This is open source data platform which powers official open-data portal for governments and public organizations, including India's data.gov.in
url :- Dataset1 link ,Dataset2 link

DATA SCIENCE WORKFLOW

DATA COLLECTION	CLEANING & PREPROCESSING	EDA & VISUALIZATION	FORECASTING MODEL	DASHBOARD & INSIGHTS

TECHNOLOGY

Python • Pandas • NumPy • Matplotlib • Seaborn
Scikit-learn • Jupyter Notebook • Interactive Dashboard

EXPECTED OUTCOME

The project will deliver an interactive electricity intelligence dashboard showing historical trends, renewable-energy growth, state-wise comparisons, energy-mix analysis, and future generation forecasts.

HISTORICAL DATA → INSIGHTS → COMPARISONS → FORECASTS

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